

11.1

$$3. (a) f(1,1,1) = \sqrt{1} + \sqrt{1} + \sqrt{1} + \ln(4 - 1^2 - 1^2 - 1^2) \\ = 3$$

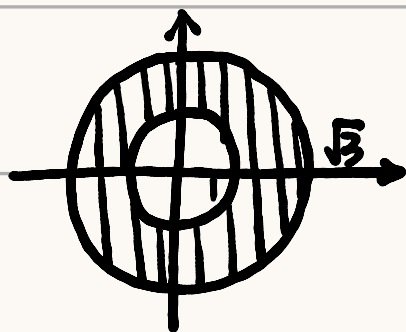
$$(b) \ x, y, z \geq 0 \quad 4 - x^2 - y^2 - z^2 > 0 \Rightarrow x^2 + y^2 + z^2 < 4 \\ \text{thus, the domain: } \begin{cases} x^2 + y^2 + z^2 < 4 \\ x, y, z \geq 0 \end{cases}$$

$$4. (a) g(1,2,3) = 1^3 \cdot 2^2 \cdot 3 \sqrt{10 - 1 - 2 - 3} = 24$$

$$(b) \ 10 - x - y - z \geq 0 \Rightarrow x + y + z \leq 10$$

$$\text{thus, the domain: } x + y + z \leq 10$$

$$10. -1 \leq x^2 + y^2 - 2 \leq 1 \Rightarrow 1 \leq x^2 + y^2 \leq 3$$



$$\text{thus, the domain } x^2 + y^2 \in [1, 3]$$

$$21. f(-3,3) \approx 55 \quad f(3,-2) \approx 35, \text{ the shape likes a CONE}$$

22. I: g , a paraboloid: the density near the origin is sparse, the outer's density is dense, because the slope at the bottom of paraboloid is smaller than other places.

II: f a cone: the density is same because

the slope of cone is a straight line.

$$29. \quad z = ye^x \Rightarrow y = \frac{z}{e^x}$$

$$32. \quad f(x,y) = \frac{y}{x^2 + y^2} = z$$

$$35. \quad T(x,y) = \frac{100}{1 + x^2 + 2y^2} = z \Rightarrow y = \pm \sqrt{\frac{50}{z} - \frac{1}{2} - \frac{x^2}{2}}$$

52. (a) move 2 units towards right

(b) move 2 units downwards

(c) move 3 units towards left
move 4 units upwards

41: C II 42: A IV 43: F I

44: E III 45: B VI 46: D V

11.2

2. (a) continuous the temperature is continuous

(b) discontinuous there might be cliff

(c) discontinuous might be floor function

$$6. \ y = kx \quad \frac{5y^4 \cos^2 x}{x^4 + y^4} = \frac{5k^4 x^4 \cos^2 x}{x^4 + k^4 x^4} = \frac{5k^4 \cos^2 x}{1 + k^4} \text{ not exist}$$

$$7. \ y = kx \quad \frac{y^2 \sin^2 x}{x^4 + y^4} = \frac{k^2 x^2 \sin^2 x}{x^4 + k^4 x^4} = \frac{k^2 \sin^2 x}{x^2(1 + k^4)} = \left(\frac{\sin x}{x}\right)^2 \frac{k^2}{1 + k^4}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \text{so } \lim_{(x,y) \rightarrow (0,0)} \frac{y^2 \sin^2 x}{x^4 + y^4} \text{ not exist}$$

$$10. \ y = kx \quad \frac{x^2 \sin^2 y}{x^2 + 2y^2} = \frac{x^2 \sin^2 kx}{x^2 + 2k^2 x^2} = \frac{\sin^2 kx}{1 + 2k^2} = 0 \text{ exist}$$

$$11. \ y = kx^2 \quad \frac{x^2 y e^x}{x^4 + 4y^2} = \frac{k x^4 e^x}{x^4 + 4k^2 x^4} = \frac{k e^x}{1 + 4k^2} \text{ not exist}$$

$$15. \ x = az^2 \quad y = bz^2 \quad \frac{xy + yz^2 + xz^2}{x^2 + y^2 + z^4} = \frac{abz^4 + bz^4 + az^4}{a^2 z^4 + b^2 z^4 + z^4} \\ = \frac{ab + b + a}{a^2 + b^2 + 1} \text{ not exist}$$

$$17. \ y = kx \quad \frac{2x^2 + 3xy + 4y^2}{3x^2 + 5y^2} = \frac{2x^2 + 3kx^2 + 4k^2 x^2}{3x^2 + 5k^2 x^2} \\ = \frac{2 + 3k + 4k^2}{3 + 5k^2} \text{ not exist}$$

$$25. \ -1 \leq x^2 + y^2 + z^2 \leq 1, \ x^2, y^2, z^2 \geq 0$$

$$\therefore x^2 + y^2 + z^2 \in [0, 1]$$

$$28. \ y = kx \quad \frac{xy}{x^2 + xy + y^2} = \frac{kx^2}{x^2 + kx^2 + k^2 x^2} = \frac{k}{1 + k + k^2} \text{ not exist}$$

$$\{(x,y) \in \mathbb{R}^2 \mid (x,y) \neq (0,0)\}$$

$$30. \ \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad (x^2 + y^2) \ln(x^2 + y^2) = r^2 \ln r^2 = 2r^2 \ln r$$

$$\lim_{r \rightarrow 0^+} 2r^2 \ln r = \lim_{r \rightarrow 0^+} \frac{2 \ln r}{\frac{1}{r^2}} = \lim_{r \rightarrow 0^+} \frac{\frac{2}{r}}{-2r^{-3}} = \lim_{r \rightarrow 0^+} -r^2 = 0$$